# ELI — Architecture Ascii

ELI v2.5.11

October 2026

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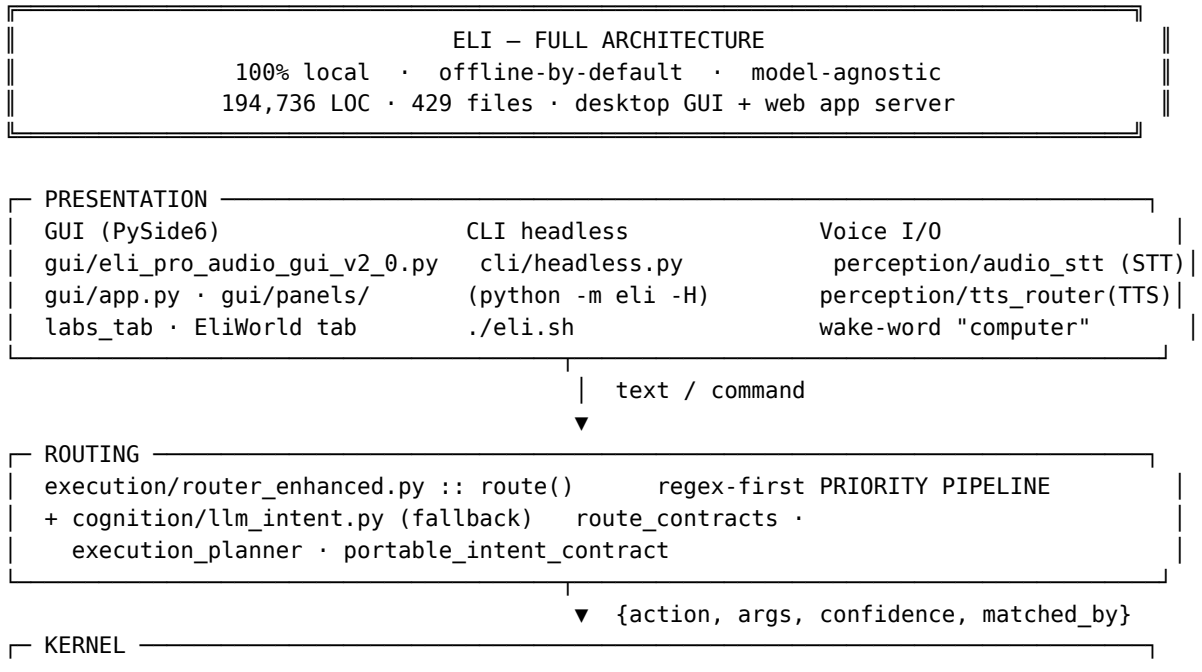

## Blueprint — ELI Full Architecture (ASCII)

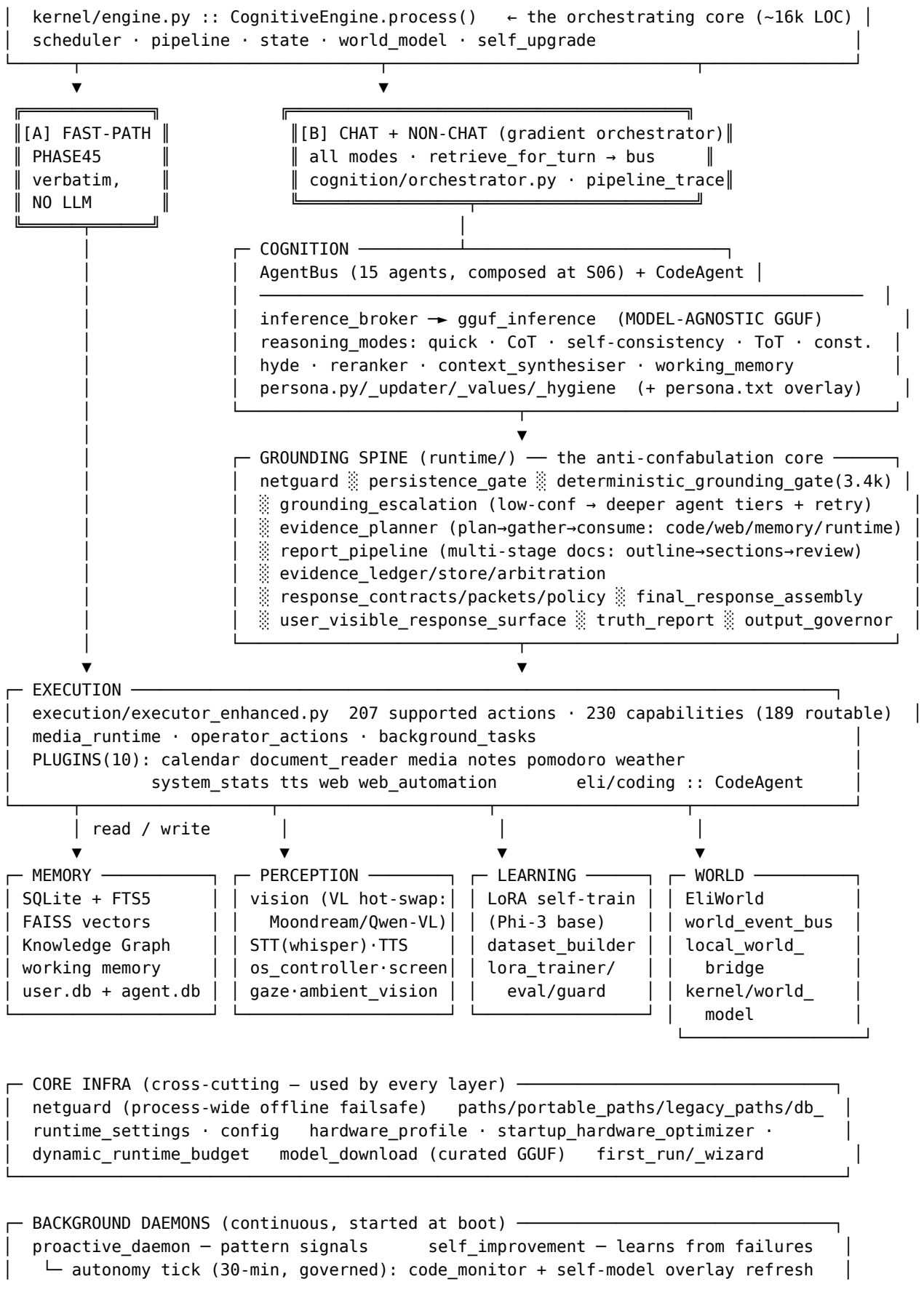
**Updated for v2.4.84.** All CHAT modes → gradient orchestrator; bus composed at S06.

The entire system in one drawing, plus the module tree and data layout. Grounded in the real source (see `architecture.md` for prose, `diagrams.md` for the pipeline/memory/gating close-ups). Every layer and box maps to a real path.

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### A. Full system — layered block diagram




kernel/engine.py :: CognitiveEngine.process()  ← the orchestrating core (~16k LOC)
scheduler · pipeline · state · world_model · self_upgrade
[A] FAST-PATH
PHASE45
verbatim,
NO LLM
[B] CHAT + NON-CHAT (gradient orchestrator)
all modes · retrieve_for_turn → bus
cognition/orchestrator.py · pipeline_trace
COGNITION
AgentBus (15 agents, composed at S06) + CodeAgent
inference_broker → gguf_inference  (MODEL-AGNOSTIC GGUF)
reasoning_modes: quick · CoT · self-consistency · ToT · const.
hyde · reranker · context_synthesiser · working_memory
persona.py/_updater/_values/_hygiene  (+ persona.txt overlay)
GROUNDING SPINE (runtime/) ── the anti-confabulation core
netguard ▒ persistence_gate ▒ deterministic_grounding_gate(3.4k)
▒ grounding_escalation (low-conf → deeper agent tiers + retry)
▒ evidence_planner (plan→gather→consume: code/web/memory/runtime)
▒ report_pipeline (multi-stage docs: outline→sections→review)
▒ evidence_ledger/store/arbitration
▒ response_contracts/packets/policy ▒ final_response_assembly
▒ user_visible_response_surface ▒ truth_report ▒ output_governor
EXECUTION
execution/executor_enhanced.py  207 supported actions · 230 capabilities (189 routable)
media_runtime · operator_actions · background_tasks
PLUGINS(10): calendar document_reader media notes pomodoro weather
system_stats tts web web_automation
eli/coding :: CodeAgent
read / write
MEMORY
SQLite + FTS5
FAISS vectors
Knowledge Graph
working memory
user.db + agent.db
PERCEPTION
vision (VL hot-swap:
Moondream/Qwen-VL)
STT(whisper)·TTS
os_controller·screen
gaze·ambient_vision
LEARNING
LoRA self-train
(Phi-3 base)
dataset_builder
lora_trainer/
eval/guard
WORLD
EliWorld
world_event_bus
local_world_
bridge
kernel/world_
model
CORE INFRA (cross-cutting — used by every layer)
netguard (process-wide offline failsafe)  paths/portable_paths/legacy_paths/db_
runtime_settings · config  hardware_profile · startup_hardware_optimizer ·
dynamic_runtime_budget  model_download (curated GGUF)  first_run/_wizard
BACKGROUND DAEMONS (continuous, started at boot)
proactive_daemon — pattern signals
self_improvement — learns from failures
└─ autonomy tick (30-min, governed): code_monitor + self-model overlay refresh

```
│      + goal/scheduler ticks → proposals (observe-only / memory-write; need approval)│
│  habits_scheduler/habits — routines      scheduler — jobs                            │
│  background_tasks — async heavy work     reflection loop   ambient_vision loop       │
│  scheduled_tasks (durable overnight/timed)   world_event_bus ← confidence events     │
└──────────────────────────────────────────────────────────────────────────────────────┘

          OUTPUT  ←── output_governor / sanitiser ←── inference ←── (any path)
                        │
                        └─► TTS (Piper)  /  GUI render
```

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## B. The 15 agents

```
AGENT BUS  (cognition/agent_bus.py :: AgentBus.dispatch → DispatchResult)
selects a set → runs on a dependency DAG → aggregates grounding_confidence
┌──────────────────────┬──────────────────────────────────────────────────────────┐
│ BusMemoryAgent       │ recall: SQLite/FTS5/FAISS → context                      │
│ KnowledgeGraphAgent  │ entities/relations from the KG                           │
│ SystemAgent          │ runs executor actions (WEB_SEARCH, RUNTIME_AUDIT, …)     │
│ OrchestratorAgent    │ plan / coordinate grounded synthesis                     │
│ FileCodeAgent        │ searches the codebase for code-grounded answers          │
│ IntrospectionBus…    │ live runtime + gathers identity/awareness audits as      │
│                      │ evidence (persona summarises — never a data dump)        │
│ CapabilityAgent      │ capability manifest (what ELI can do)                    │
│ ReflectionAgent      │ session reflection / insights                            │
│ ProactiveAgent       │ proactive patterns / signals                             │
│ HabitAgent           │ learned habits                                           │
│ SelfImprovementAgent │ self-improvement state                                   │
│ FrontierAgent        │ frontier reasoning hooks                                 │
│ PluginAgent          │ dispatch to registered plugins                           │
│ VoiceAgent           │ STT/TTS engine state                                     │
│ CriticAgent          │ second-tier verifier: checks retriever agents' output    │
├──────────────────────┼──────────────────────────────────────────────────────────┤
│ + CodeAgent          │ SEPARATE pipeline: plan→search→verify→repair             │
│   eli/coding/agent   │ (CODE_SOLVE / GENERATE_SCRIPT) — not a bus agent         │
└──────────────────────┴──────────────────────────────────────────────────────────┘
```

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## C. Module tree (LOC · key files · role)

```
eli/  (194,736 LOC, 429 files)  ·  api/server.py  (FastAPI web app + dashboard)
│
├── __main__.py ·····················  entry dispatch (GUI | --headless)
│
├── kernel/              17.5k ─── the core
│   ├── engine.py        16.0k     CognitiveEngine.process() — the spine
│   ├── scheduler.py               timed jobs
│   ├── pipeline.py · state.py · world_model.py · self_upgrade.py
│
├── execution/           26.7k ─── route → act
│   ├── executor_enhanced.py   16.0k   207 supported / 230 manifest
│   ├── router_enhanced.py      8.4k   priority pipeline
│   ├── execution_planner.py · route_contracts.py
│   ├── operator_actions.py · operator_policy.py
```

```
│   ├── media_runtime.py
│   └── portable_intent_contract.py
│
├── cognition/         20.8k ── think
│   ├── agent_bus.py    3.5k   15 agents + dispatch
│   ├── orchestrator.py        12-stage deep retrieval
│   ├── gguf_inference.py 3.2k · inference_broker.py   model-agnostic inference
│   ├── reasoning_modes.py · hyde.py · reranker.py · llm_intent.py
│   ├── persona.py/_updater/_values/_status/_hygiene  (+ persona.txt, persona.auto.txt)
│   ├── context_synthesiser.py · working_memory.py
│   ├── output_governor.py · response_governance.py · response_sanitizer.py
│   └── grounded_status.py · introspection_agent.py · tone_analyzer.py
│
├── runtime/           35.4k ── grounding spine + daemons (95 files)
│   ├── deterministic_grounding_gate.py 3.4k · grounding_escalation.py
│   ├── evidence_ledger/arbitration.py
│   ├── persistence_gate.py · truth_report.py · control_contracts.py
│   ├── response_contracts/packets/policy.py · final_response_assembly/provider.py
│   ├── user_visible_response_surface.py · personal_memory_*.py · reflection.py
│   ├── background_tasks.py · self_improvement.py · code_monitor.py
│   ├── capability_sync.py · pending_proposal.py · runtime_policy.py
│   └── agenda.py (calendar + reminders)
│
├── memory/             8.8k ── remember (11 files)
│   ├── memory.py       5.7k   Memory · SQLite + FTS5 · storage policy in policy.py
│   └── vector_store.py        FAISS index
│
├── perception/        10.2k ── sense (23 files)
│   ├── vision.py · analyze_image/csv/pdfs/mesh.py · ambient_vision.py
│   ├── audio_stt.py · local_whisper_stt.py · voice_worker_streaming.py
│   ├── tts_router.py · os_controller.py · screen_locator.py · gaze_engine.py
│
├── planning/           4.0k ── proactivity (19 files)
│   ├── proactive_daemon.py · habits_scheduler.py · habits.py · jobqueue_cli.py
│
├── coding/             2.1k ── CodeAgent (plan→search→verify→repair, 12 files)
│
├── learning/           4.3k ── LoRA self-training (14 files)
│   ├── lora_trainer/eval/guard.py · dataset_builder/filters.py
│   ├── bootstrap_phi3_base.py · base_model_resolver.py · training_preflight.py
│
├── plugins/            5.9k ── manager + bundled plugins (34 files)
│   └── calendar · document_reader · media · notes · pomodoro · weather ·
│       system_stats · tts · web · web_automation
│
├── world/              1.8k ── EliWorld (world_event_bus, local_world_bridge)
│
├── core/              11.2k ── infra
│   ├── netguard.py            offline failsafe + allow_network()
│   ├── paths.py · portable_paths.py · legacy_paths.py · db_paths.py
│   ├── runtime_settings.py · config.py · grounding.py
│   ├── hardware_profile.py · startup_hardware_optimizer.py · dynamic_runtime_budget.py
│   ├── model_download.py
│
├── gui/               27.4k ── PySide6 desktop
```

```
│   ├── eli_pro_audio_gui_v2_0.py 13.1k · app.py · labs_tab.py 5.7k
│   └── panels/  (startup.py: model picker + FirstBootWizard, HardwareTuningDock)
│
├── tools/              7.6k ─── image_engine · news · document tools
├── contracts/ 0.8k · cli/ 0.1k · system/ 1.2k · utils/ 1.8k · setup/ 1.9k · onboarding/ 0.7k
```

---

## D. On-disk data layout

```
artifacts/
├── db/
│   ├── user.sqlite3      memories · memories_archive · memory_meta · semantic ·
│   │                     conversation_turns · conversations · session_summaries ·
│   │                     kg_entities · kg_relations · recall_log · runtime_events ·
│   │                     learning_replay · observations · habits/habit_events/habit_rules ·
│   │                     user_patterns · user_model · eli_stances · belief_revisions ·
│   │                     news_articles · news_reflections   (+ FTS5 indexes)
│   ├── agent.sqlite3     agent_dispatches · agent_metrics · improvements · failures ·
│   │                     corrections · capability_proposals · error_tracking
│   ├── system_index.sqlite3   OS index
│   └── coding_memory.sqlite3  coding_bug_fixes
├── vectors/index.faiss   semantic index   (embedder: nomic-embed-…Q4_K_M.gguf)
├── conversations/
├── runtime/ · runtime_snapshot.json        live model/runtime truth
├── world/{snapshots,ledger,journal}/
├── documents/ · scripts/ · analyze_image_*/
│
config/   settings.json (gitignored) · templates/settings.template.json ·
          settings.example.json · plugins_state.json
models/   <your>.gguf · embeddings/ · whisper/ · image/     tts_piper/  Piper voices
```

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## E. The flow in one line

```
INPUT → ROUTER → ENGINE ─┬─ FAST-PATH ──────────────────────────► OUTPUT (verbatim)
                         └─ ORCHESTRATOR (all CHAT modes, gradient)
                                ├─ retrieve_for_turn() ──► MEMORY
                                ├─ dispatch_specialists() ──► 15-agent BUS
                                └─ broker.infer() → finalize_turn() (S12)
   GATES: netguard · persistence · grounding · escalation · governor
   INFERENCE (broker → gguf, model-agnostic) → OUTPUT → TTS/GUI
   BACKGROUND: proactive · self-improve · habits · learning(LoRA) · world · scheduler
```

Companion docs: `architecture.md` (prose, every subsystem) · `diagrams.md` (pipeline / memory / gating close-ups) · `tools/eval/README.md` (the measurement layer). "'